ZCAS University

CCS1501 ENGINEERING MATHEMATCIS

DEFERRED EXAMINATION

TUESDAY, 19TH JUNE, 2026

TIME: 14:00 TO 17:00 HOURS

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer Three (3) questions from this section.
3. This question paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Question one

(a) Given the set $P = \{x: 0 \leq x \leq 3, x \in Z\}$ and $Q = \{x: 2 < x \leq 4,\ x \in Z\}$, where P and Q are subsets of $M = \{x: 0 \leq x \leq 10, x \in Z\}$. List

(i) $P \cap Q$ **[2 Marks]**

(ii) $Q \cup P$ **[2 Marks]**

(iii) $P \times Q$ **[2 Marks]**

(b) Solve the given system of equations $\begin{cases} 2\ log_x y = 1 \\ xy = 64 \end{cases}$ **[4 Marks]**

(c) The remainder upon dividing the polynomial $\alpha x^4 + \beta x^3 - 8x^2 + 6$ by $x^2 - 1$ is $2x + 1$. Compute the value of α and β. **[4 Marks]**

(d) (i) Express as a single logarithm $2\log_4(2x + 3) - \log_4 x - \log_4(2x - 1)$ **[1 Mark]**

(ii) Hence, solve the equation $2\log_4(2x + 3) = 1 + \log_4 x + \log_4(2x - 1)$ **[3 Marks]**

(e) Solve the equation $3^{2x+1} + 9 = 3^{x+3} + 3^x$ **[5 Marks]**

(f) Rationalize the denominator of $\frac{1}{2\sqrt{2}-\sqrt{3}}$ **[3 Marks]**

(f) (i) Write down the first three terms in the Binomial expansion of $\sqrt{1+x}$ **[4 Marks]**

(ii) Hence, estimate the value of $\sqrt{1.08}$ correct to four decimal places. **[5 Marks]**

(g) Resolve $\frac{x^3+3x^2-12x+36}{x^4-16}$ into partial fractions. **[5 Marks]**

[TOTAL: 40 Marks]

SECTION B: Attempt any THREE questions in this section

Question two

(a) Solve for x(i) $|x - 2| \leq 15$ **[4 Marks]**

(b) Given the polynomial $P(x) = x^3 + 4x^2 + x - 6$

(i) Factorize $P(x)$ completely **[3 Marks]**

(ii) Hence, resolve $\frac{5x+19}{x^3+4x^2+x-6}$ into partial fractions **[3 Marks]**

(c) Find the value of n if the coefficient of the second term in the expansion of

$(1 - x)(1 + 2x)^n$ is 19. **[3 Marks]**

(d) Given that $f(x) = ax^3 + (a + b)x^2 + (a + 2b)x + 1$ is exactly divisible by $(x + 1)$.

(i) Express b in terms a. **[2 Marks]**

(ii) Use synthetic division to find the **quotient** when $f(x)$ is divided by $(x + 1)$. **[3 Marks]**

(iii) If the quotient in part (ii) has complex roots, show that $a^2 - 6a + 1 < 0$. **[2 Marks]**

[Total: 20 Marks]

Question three

(a) Solve for x $\ln(x + 2) = \ln(x - 4) + \ln(3)$ **[4 Marks]**

(b) (i) Determine whether the function $h(x) = x^3 + x$ is odd, even or neither **[2 Marks]**

(ii) Given the functions $f(x) = \frac{x+3}{x-1}, where\ x \neq 1$ and $g(x) = 2x + 1$, verify that $f^{-1}(2) = fog\left(\frac{1}{2}\right)$ **[4 Marks]**

(c) The expression $\frac{3x+7}{x^3+x^2-2x}$ can be resolved into sums of partial fractions by firstly finding the factors of its denominator. Determine these factors. **[5 Marks]**

(d) (i) Define a one-to-one function **[2 Marks]**

(ii) Show that $f(x) = 9x - 2$ is a one to one function **[3 Marks]**

[Total: 20 Marks]

Question four

(a) Show that the roots of a quadratic equation $ax^2 + bx + c = 0$ **[5 Marks]**

(b) (i) If $\alpha\ and\ \beta$ are roots to the equation $2x^2 + 3x + 2 = 0$, find the quadratic equation whose roots are $\frac{1}{\alpha\beta}\ and\ \frac{1}{\alpha+\beta}$. **[5 Marks]**

(ii) If $\log x - 2\log y = 1$ and $xy = 80$. Find the values of $x\ and\ y$. **[5 Marks]**

(c) Find the term independent of x in the expansion of $\left(x - \frac{1}{x^2}\right)^9$ **[5 Marks]**

[Total: 20 Marks]

Question five

(a) Factorize completely $x^3 - 6x^2 + 11x - 6$ **[5 Marks]**

(b) Let $= \begin{pmatrix} 1 & 0 & 2 \\ 1 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix}$, find

(i) A^2 **[3Marks]**

(ii) A^{-1} **[7 Marks]**

(ii) Hence or otherwise, solve the following system of equations

$$x \quad + 2z = 1$$
$$x + y + 2z = 2$$
$$y + z = 3$$

[5 Marks]

[TOTAL: 20 Marks]

END OF EXAMINATION